

Holy Faith Academy, Dhuliyan, M.S.D.

Affiliated to CICSE, New Delhi

Class: IX
Semester-1

Subject: Computer

F.M: 100
Time Allowed: One and a Half Hours

Instructions:

- Answers to this paper must be written on the paper provided separately.
 - You will not be allowed to write during the first 10 minutes. This time is to be spent on reading the question paper.
 - The time given at the head of this paper is the time allowed for writing the answers.
 - Attempt all questions from Section A and any four questions from Section B.
 - The intended marks for questions or parts of questions are given in brackets [].
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1 Section A (40 Marks)

(Attempt all questions.)

1.1 Question 1 [10]

- a) What is a program?
- b) List the OOP concepts?
- c) What is meant by object code?
- d) What is meant by a class?
- e) Distinguish between the terms "B" and "b".

1.2 Question 2 [20]

- a) Explain explicit type conversion with one example.
- b) Which of the following is not a Token?
 - i. Operator
 - ii. Statement
 - iii. Keyword
 - iv. Identifier
- c) The storage size of char data type is
 - i. 2 bytes

- ii. 3 bytes
 - iii. 1 byte
 - iv. None of these
- d) The entity which has same characteristics and behavior is called
- i. class
 - ii. Object
 - iii. abstraction
 - iv. Polymorphism
- e) Which of the following entity is a blueprint of an object?
- i. abstraction
 - ii. Object
 - iii. class
 - iv. Polymorphism
- f) Which of the following is a primitive data type?
- i. class
 - ii. Structure
 - iii. Boolean
 - iv. Bitwise
- g) The new Line character is represented by
- i. \n
 - ii. 'a'
 - iii. \t
 - iv. None of these
- h) Which of the following is not an unary operator?
- i. ++
 - ii. -
 - iii. ++
 - iv. -
- i) The expression $19\%3$ gives the output
- i. 2
 - ii. 1
 - iii. 3.66

- iv. None of these
- j) The value of y is 5. Then the value stored in variable m after evaluation of $m = y++ + 10$ is
 - i. 15
 - ii. 16
 - iii. 5
 - iv. None of these

1.3 Question 3 [10]

- a) What are the rules of naming identifiers?
- b) What is the type conversion in Java? Explain with example.
- c) Why keywords are also called reserved words? Explain.
- d) Evaluate the expression when x is 4:
`x += x++ * ++x % 2;`
- e) Write the value of n after execution:
`char ch = 'd';`
`int n = ch + 5;`

2 Section B (60 Marks)

(Attempt any four questions.)

2.1 Question 4 [10]

Write a program in Java that inputs a number. Now check and display whether the inputted number is Even or Odd. Give suitable message with your answer.

2.2 Question 5 [10]

Write a program in Java that inputs the length and breadth of the rectangle. Now calculate and display the area and perimeter of the rectangle with suitable message.

2.3 Question 6 [10]

Write a program in Java that inputs a number. Now calculate and display the following with suitable message:

- a) The value of the original number inputted.
- b) Double of the inputted number.
- c) Triple of the inputted number.
- d) Square of the inputted number.
- e) Square root of the inputted number.

2.4 Question 7 [10]

Write a program in Java that inputs the value of the two numbers. Now calculate and display the following with suitable message:

- a) Sum of the two inputted numbers
- b) Difference of the two inputted numbers
- c) Product of the two inputted numbers
- d) Quotient when 1st number is divided by 2nd number.
- e) Remainder obtained when 1st number divided by 2nd number.

2.5 Question 8 [10]

Write a program in Java that inputs the radius of a circle. Now calculate and display the following with suitable message:

- i) Diameter of the circle
- ii) Area of the circle
- iii) Perimeter of the circle

2.6 Question 9 [10]

Write a program in Java that inputs cost price and Selling price of an article. (Assuming selling price is greater than Cost price). Now calculate and display the Profit and profit percent with suitable message.